

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

60

Automotive Front/Rear Bumper Fascia (BUMPER.stp)

1691 × 647 × 528 mm · 2059.9 cm³ · AI 5.562 kg / Steel 16.170 kg

Manufacturability: 60/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	1690.9 × 647.4 × 528.1 mm	Surface Area	16261.7 cm ²
Volume	2059.90 cm ³	Weight (AI)	5.562 kg
Weight (Steel)	16.170 kg	Weight (Plastic)	2.163 kg

§2 — Detected Features

Feature Type	Count	Significance	Description
Undercut	4	High	4 undercut faces relative to Z draw direction — require side-actions or core-pulls
Zero-draft face	2	Medium	2 faces with 0° draft — high ejection drag / scuffing risk

§3 — Material Analysis

AI-suggested

Primary Material: Unspecified (50% confidence)
undefined

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
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§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£0.00	£0.00	£0.00

Net Weight	5.562 kg	Material	—
Recommended Process	machining	Cycle Time	—
Setup Time	—	Operations	0 ops

Operations Detail

Process-Specific Parameters

Cavities	—
Wall Thickness	2.5 mm
Mould Cost	—
Mould Life	— shots
Projected Area	10946.8 cm²

§8 — AI Analysis Explanation